Fisker Ocean ICC CAN gateway (IBUS1/IBUS2 man-in-the-middle)
Teensy 4.1 + MCP2518FD, TJA1043 (car side) / TJA1051T/3 (ICC side), relay bypass

Connectors



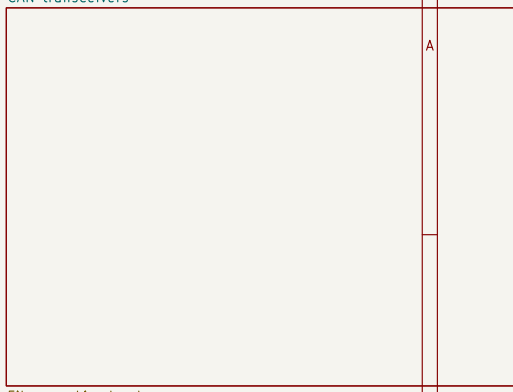
File: connectors.kicad_sch

Power



File: power.kicad_sch

CAN transceivers



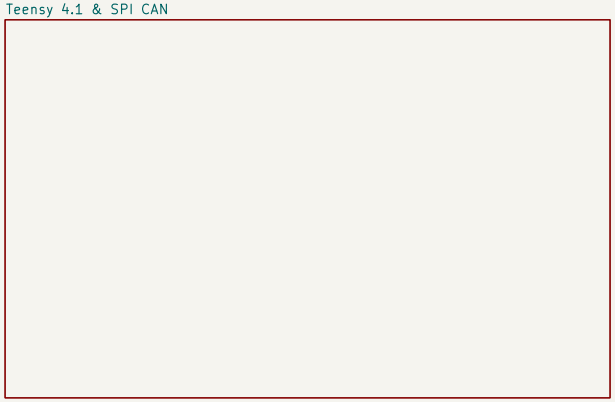
File: can.kicad_sch

Bypass relays & termination



File: bypass.kicad_sch

Teensy 4.1 & SPI CAN



File: mcu.kicad_sch

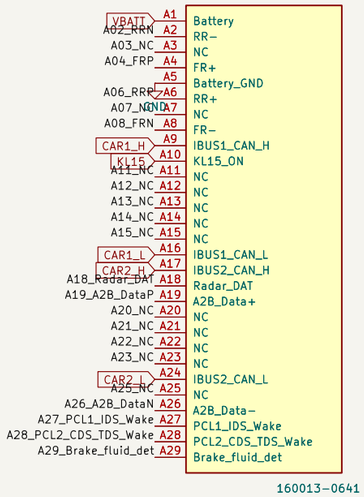
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Title:

Size: A4	Date:	**Rev:**
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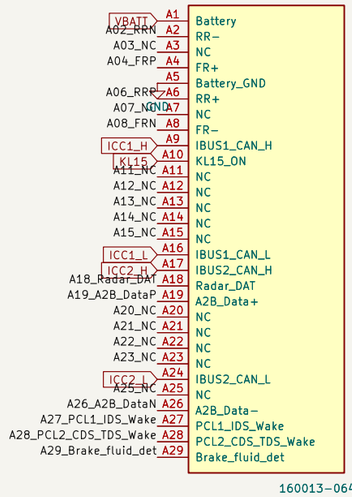
J1 = car harness (stock receptacles plug in here) J2 = jumper harness to the ICC (two 160014 receptacle sets)
Everything passes straight through except IBUS1/IBUS2 CAN (A9/A16, A17/A24); Battery A1 / GND A5 / KL15 A10 also feed the gateway.

J1A



160013-0641

J2A

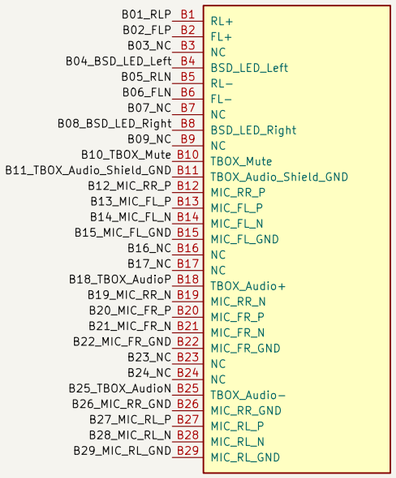


160013-0641

PWR_FLAG VBATT

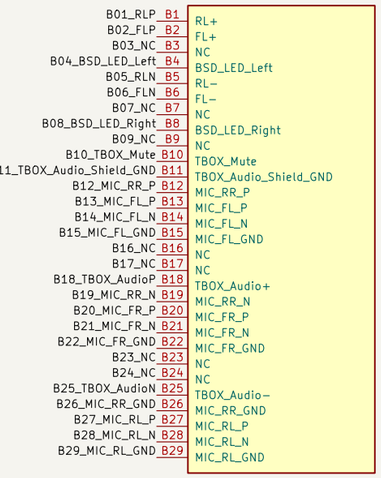
PWR_FLAG GND

J1B



160013-0641

J2B



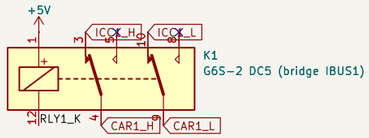
160013-0641

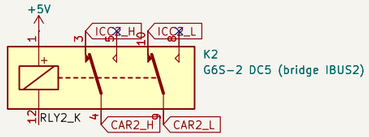
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File: connectors.kicad_sch

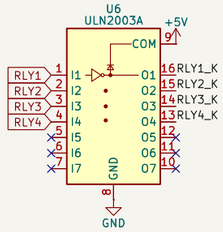
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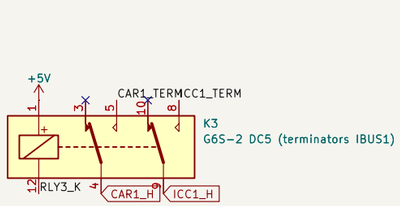
Size: A3	Date:	**Rev:**
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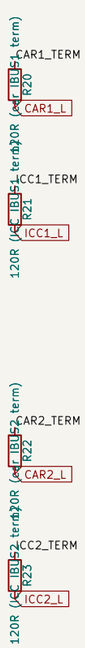
C10
100nF (U2 VCC)
C11
100nF (U2 VIO)
C12
100nF (U2 VBAT)
C16
100nF (U4 VCC)
C17
100nF (U4 VIO)
U2
TJA1043T/1 (car IBUS1)
CAN1_RX
CAN1_TX
CAR1_EN
CAR1_STBN
CAR1_ERRN
CAR1_WAKE
CAR1_H
CAR1_L
INH1
D10
NUP2105L (car IBUS1 ESD)
U4
TJA1051T/3 (ICC IBUS1)
CAN3_TX
CAN3_RX
ICC1_H
ICC1_L
D12
NUP2105L (ICC IBUS1 ESD)
KL15
R10
10k (KL15 WAKE)
CAR1_WAKE
C13
100nF (U3 VCC)
C14
100nF (U3 VIO)
C15
100nF (U3 VBAT)
C18
100nF (U5 VCC)
C19
100nF (U5 VIO)
U3
TJA1043T/1 (car IBUS2)
CAN2_RX
CAN2_TX
CAR2_EN
CAR2_STBN
CAR2_ERRN
CAR2_WAKE
CAR2_H
CAR2_L
INH2
D11
NUP2105L (car IBUS2 ESD)
U5
TJA1051T/3 (ICC IBUS2)
CAN4_TX
CAN4_RX
ICC2_H
ICC2_L
D13
NUP2105L (ICC IBUS2 ESD)
KL15
R11
10k (KL15 WAKE)
CAR2_WAKE
Sheet: /CAN transceivers/
File: can.kicad_sch
Title:
Size: A3
Date:
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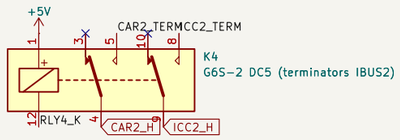












Sheet: /Bypass relays & termination/		
File: bypass.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
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PWR_FLAG
PWR_FLAG
VBAT_RTC

U7
GND
0_RX1_CRX2_CS1
1_TX1_CTX2_MISO1
2_OUT2
3_LRCLK2
4_BCLK2
5_IN2
6_OUT1D
7_RX2_OUT1A
8_TX2_IN1
9_OUT1C
10_CS_MQSR
11_MOSI_CTX1
12_MISO_MQSL
3V3
24_A10_TX6_SCL2
25_A11_RX6_SDA2
26_A12_MOSI1
27_A13_SCK1
28_RX7
29_TX7
30_CRX3
31_CTX3
32_OUT1B
D−
D+
5V
D−
D+
GND
GND
VUSB
VIN
GND
3V3
23_A9_CRX1_MCLK1
22_A8_CTX1
21_A7_RX5_BCLK1
20_A6_TX5_LRCLK1
19_A5_SCL
18_A4_SDA
17_A3_TX4_SDA1
16_A2_RX4_SCL1
15_A1_RX3_SPDIF_IN
14_A0_TX3_SPDIF_OUT
13_SCK_LED
GND
41_A17
40_A16
39_MISO1_OUT1A
38_CS1_IN1
37_CS
36_CS
35_TX8
34_RX8
33_MCLK2
R−
GND
T+
T−
LED
R+
VBAT
3V3
GND
PROGRAM
ON_OFF
VBAT_RTC
GND
CAN2_RX
CAN2_TX
RLY1
RLY2
RLY3
RLY4
CAR1_STBN
CAR1_EN
CAR1_ERRN
CAN4_INT
CAN4_CS
SPI_MOSI
SPI_MISO
CAN3_RX
CAN3_TX
LED1
LED2
LED3
CAN1_RX
CAN1_TX
KL15_SENSE
CAR2_ERRN
CAR2_EN
CAR2_STBN
SPI_SCK

U8
MCP2518FD-E/SL
SDI
SDO
CS
SCK
OSC2
OSC1
VDD
VSS
RXCAN
TXCAN
CLKO/SOF
INT
INT0/GPIO0/XSTBY
INT1/GPIO1
SPI_MOSI
SPI_MISO
CAN4_CS
SPI_SCK
XTAL2
XTAL1
CAN4_RX
CAN4_TX
CAN4_INT
+3V3
GND

Y1
20MHz 3225
XTAL1
XTAL2
GND

XTAL1
C20
15pF
GND
XTAL2
C21
15pF
GND

+3V3
C22
100nF (U8 VDD)
GND
+3V3
C23
10uF (U8 VDD)
GND

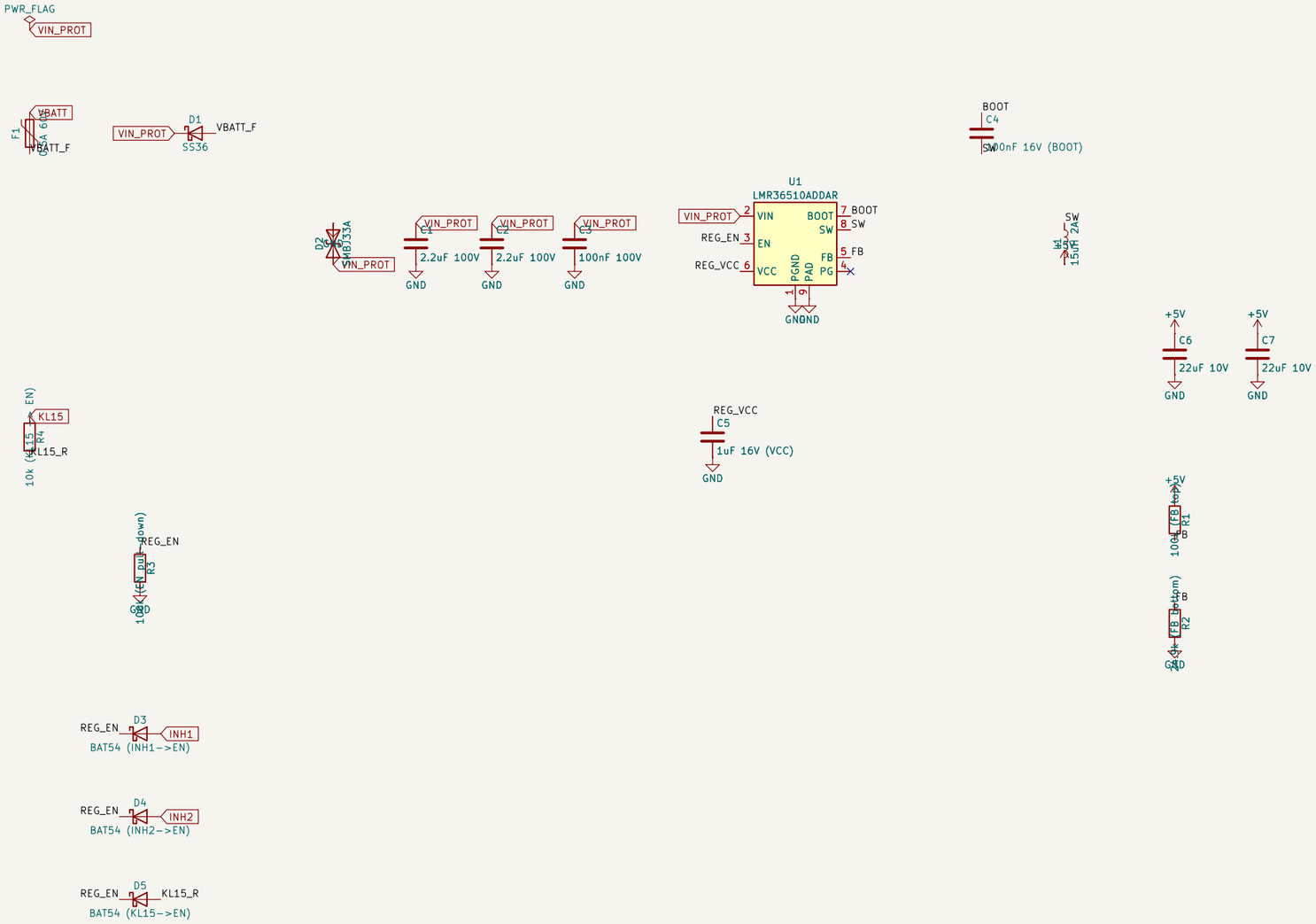
VBAT_RTC
BT1
CR2032 (RTC)
GND

KL15
R30
100k (KL15 sense)
KL15_SENSE
KL15_SENSE
R31
100k (KL15 sense)
GND
KL15_SENSE
C24
10nF (KL15 sense)
GND

LED1
R32
1k
LED1_K
LED2
R33
1k
LED2_K
LED3
R34
1k
LED3_K

D20
LED1_K
LED green (ACTIVE)
GND
D21
LED2_K
LED yellow (CAR bus)
GND
D22
LED3_K
LED blue (ICC bus)
GND

Sheet: /Teensy 4.1 & SPI CAN/
File: mcu.kicad_sch
Title:
Size: A3
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Rev:
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Sheet: /Power/
File: power.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

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